CH-2 Data Transmission

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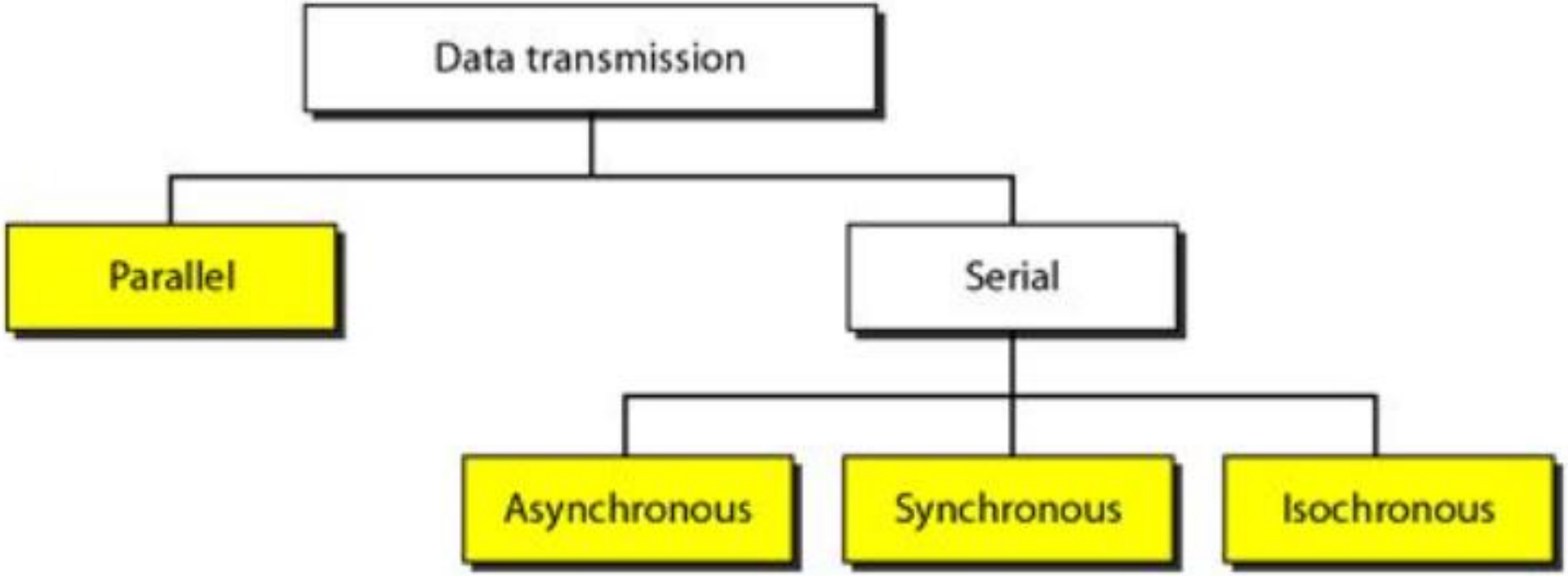
Definition of Data Transmission

Transmission is the process of sending or propagating an analog or digital signal via a medium that is wired, wireless, or fiber-optic.

- Data transmission is the **transfer of data from one digital device to another**.
- This transfer occurs via **point-to-point data streams** or channels.
- These channels may **previously** have been in the form of **copper wires** but are now much more likely to be part of a wireless network.
- Data transmission refers to the movement of data in form of bits between two or more digital devices.
- This transfer of data takes place via some form of transmission media.

Classification of Data Transmission



SERIAL TRANSMISSION

2.1 **Parallel and Serial Transmission**

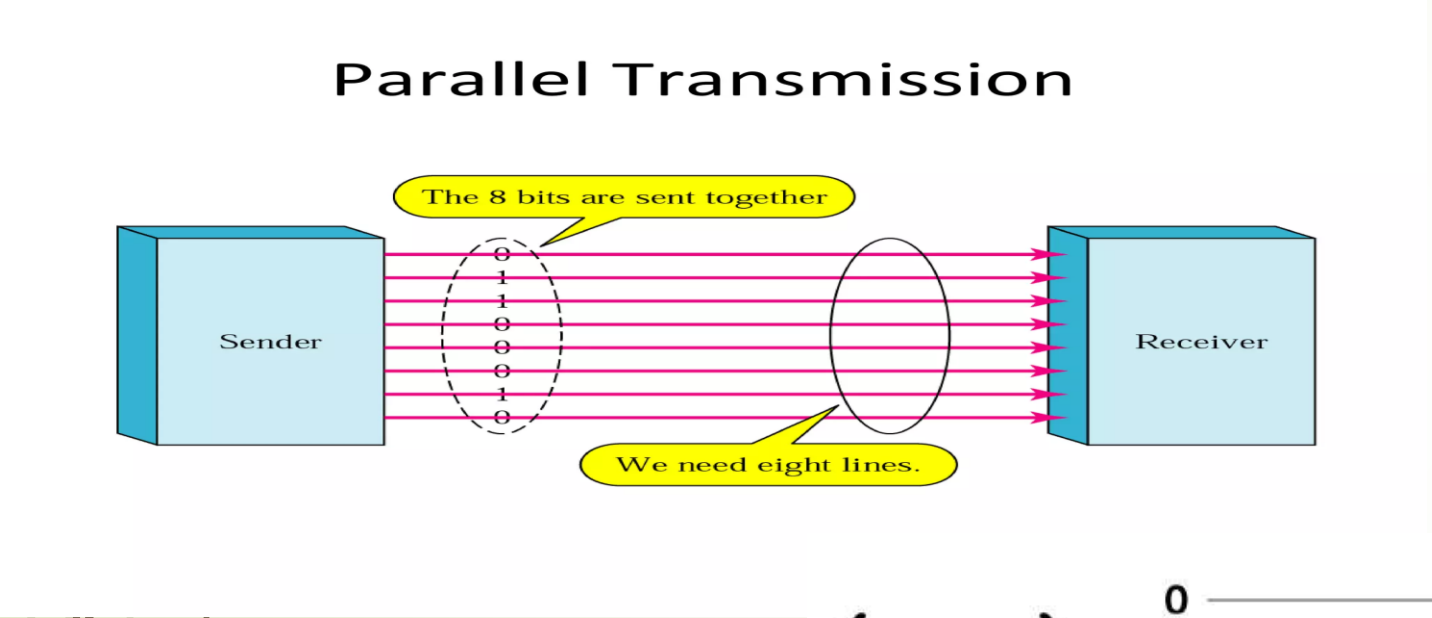
How does data transmission work between digital devices?

- Data is transferred in the form of bits between two or more digital devices.
- There are two methods used to transmit data between digital devices:
  a)**Serial transmission** and
  b)**Parallel transmission**.
- Serial data transmission sends data bits **one after another over a single channel**.
- Parallel data transmission sends **multiple data bits at the same time over multiple channels**.

2.1.1 **Parallel transmission**

- Within a communication device, the **distances between different subunits are too short**. Thus, it is normal practice to transfer data between subunits using a separate wire to carry each bit of data. There are multiple wires connecting each sub-unit **and data is exchanged using a parallel transfer mode.**
- This mode of operation results in minimal delays in transferring each word.
- In parallel transmission, all the bits of data are transmitted **simultaneously on separate communication lines.**
- In order to transmit **n** bits, **n** wires or lines are used. Thus each bit has its own line.

- All **n bits** of one group are **transmitted with each clock pulse** from one device to another i.e. **multiple bits are sent with each clock pulse**.
- Parallel transmission is used for **short distance communication**.
- As shown in the fig, eight separate wires are used to transmit 8 bit data from sender to receiver.



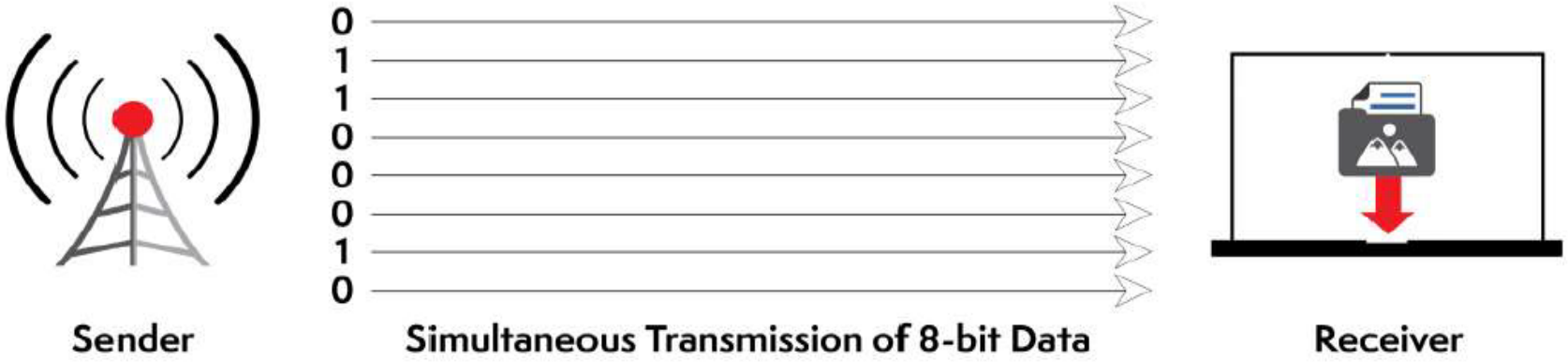


Figure : Parallel Transmission

What the figure shows

Figure 1 – Parallel Transmission (Data Received Correctly)

On the left:

- The sender transmits three 8-bit words (labeled 1, 2, and 3).
- Each row is one 8-bit binary value sent across an 8-bit parallel bus.

On the right:

- The receiver receives exactly the same 8-bit words.
- No bits are lost, changed, or mixed.

All lines arrive at the receiver in perfect synchrony, so the message prints correctly.

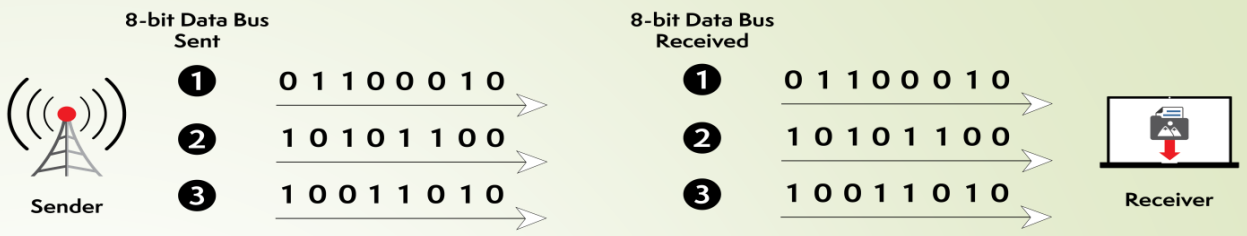


Figure 1

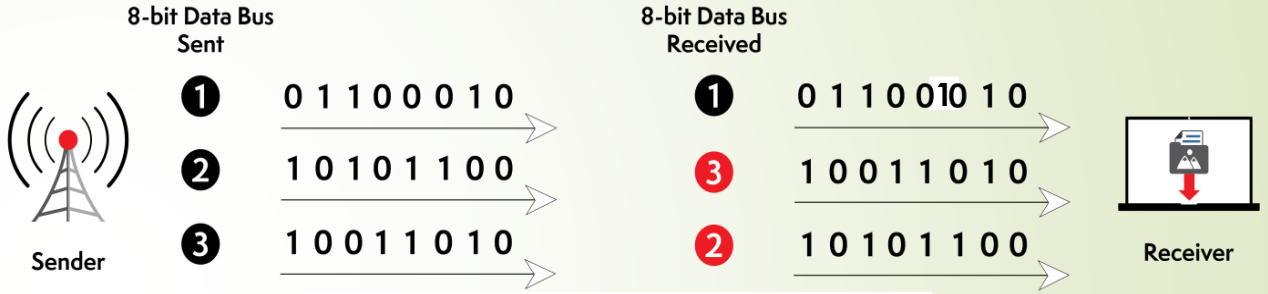


Figure 2

Figure 2 – Parallel Transmission (Data Received Incorrectly)

The sender transmits the same three 8-bit words as before.

- One or more bits become **corrupted or delayed**.
- The receiver **does not get exactly the same sequence of bits**.
- Even a **1-bit change** makes the entire word incorrect.

Happen due to:

- **Noise** on the communication lines
- **Cross-talk** between wires
- **Timing mismatch** — bits not arriving at the same moment

Advantage of parallel transmission

- It is **speedy way of transmitting data** as multiple bits are transmitted simultaneously with a single clock pulse.

Disadvantage of parallel transmission

- It is **costly method of data transmission** as it requires n lines to transmit n bits at the same time.

When is parallel transmission used to send data?

Parallel transmission is used when:

- A **large amount of data** is being sent.
- The data being sent is **time-sensitive**.
- And the data needs to be **sent quickly**.

A scenario where **parallel transmission is used to send data is video streaming**. When a video is streamed to a viewer, **bits need to be received quickly to prevent a video pausing or buffering**. Video streaming also requires the **transmission of large volumes of data**. The data being sent is also time-sensitive as slow data streams result in **poor viewer experience.**

2.1.2 Serial Transmission

- When transferring data between two physically separate devices, especially if the separation is more than a few kilometers, for reasons of cost, it is more economical to use a single pair of lines.
- Data is transmitted as a **single bit at a time using a fixed time interval** for each bit.
- In serial transmission, the various bits of data are transmitted **serially one after the other**.
- It requires **only one communication line** rather than n lines to transmit data from sender to receiver.
- Thus all the bits of data are **transmitted on single line in serial fashion**.
- In serial transmission, **only single bit is sent with each clock pulse**.
- As shown in fig., suppose an **8-bit data 01000110** is to be sent from source to destination.
- Then **least significant bit (LSB)** i,e. 0 will be transmitted first followed by other bits. The most significant bit (MSB) i.e. 0 will be transmitted in the end via single communication line.

- The internal circuitry of computer transmits data in **parallel fashion**.
- So in order to change this **parallel data into serial data**, conversion devices are used.
- These conversion devices **convert the parallel data into serial data** at the sender side so that it can be transmitted over single line.
- On receiver side, serial data received is again converted to parallel form so that the interval circuitry of computer can accept it.

- Serial transmission is used for long distance communication.

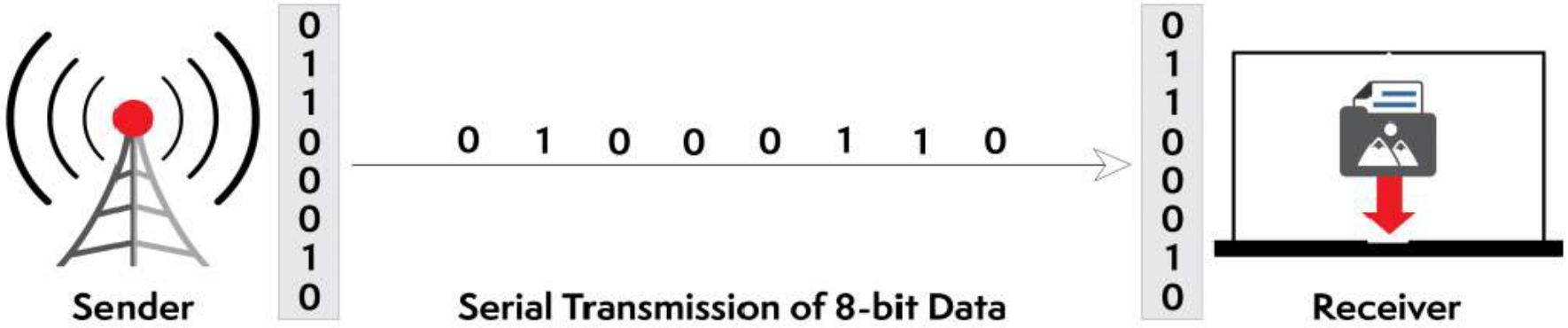
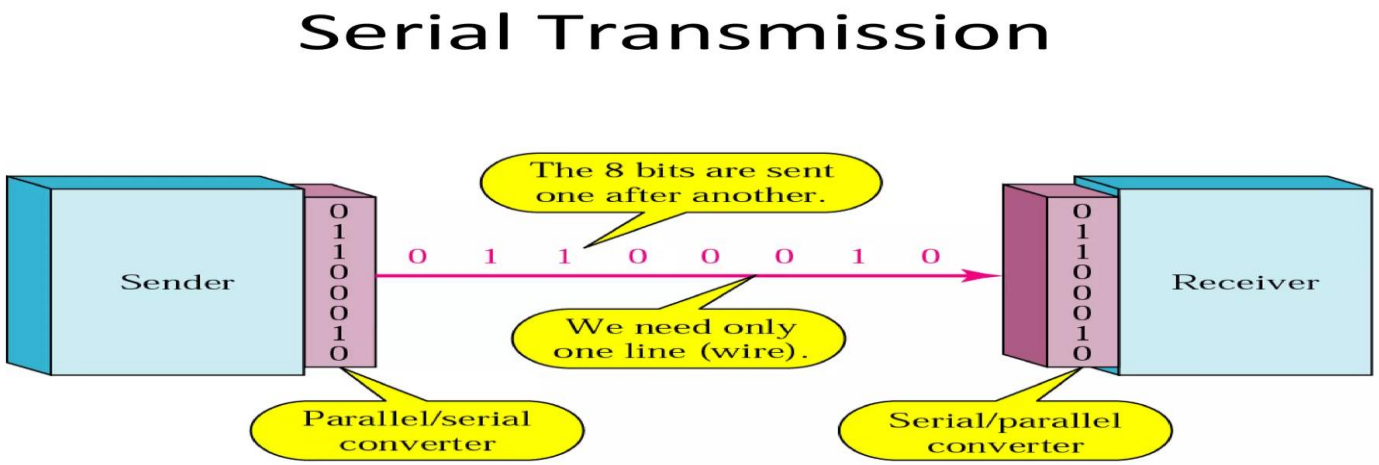


Figure 2.2: Serial Data Transmission

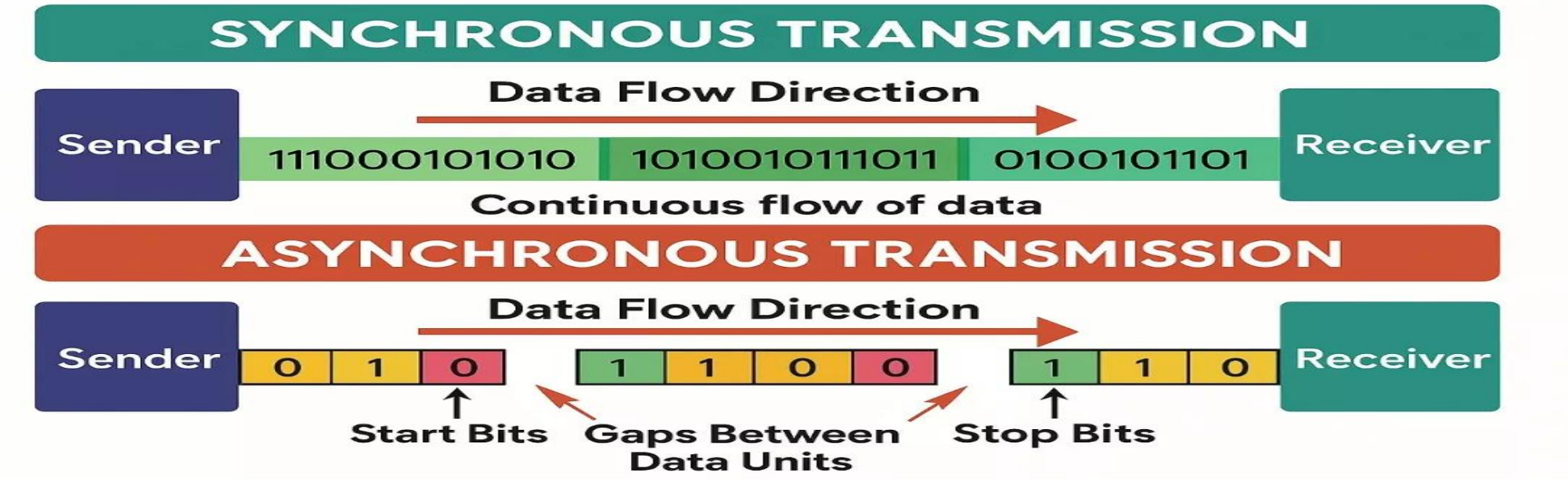


- **Advantage of Serial transmission**
- Use of single communication line **reduces the transmission line cost** by the factor of **n as compared to parallel transmission**.

- **Disadvantages of Serial transmission**
- Use of **conversion devices at source** and **destination end may lead to increase in overall transmission cost**.
- This **method is slower as compared to parallel transmission** as bits are transmitted serially one after the other.

Types of Serial Transmission

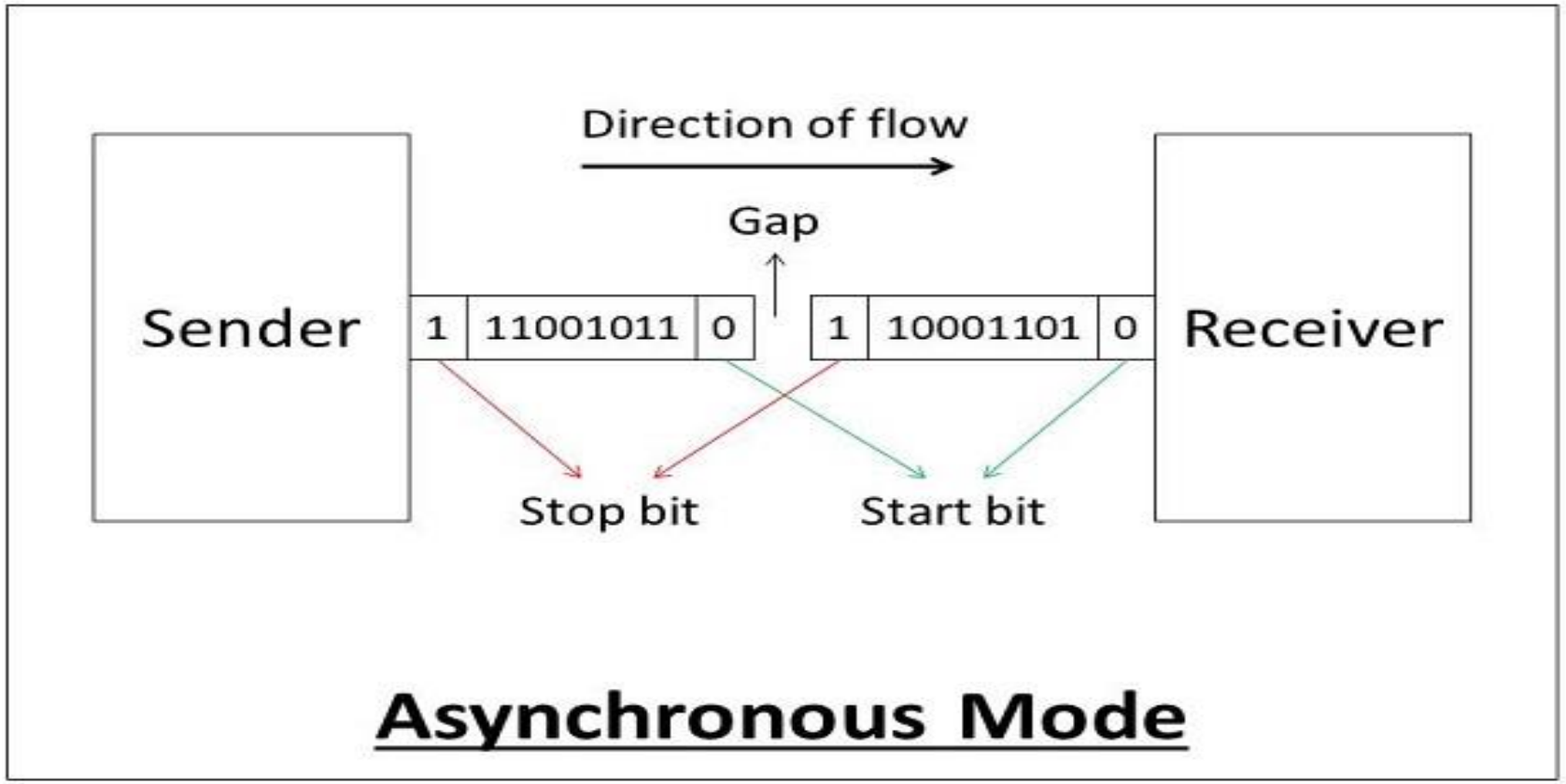
- The types of **serial transmission-synchronous ,asynchronous** and Isochronous ,these transmissions use **'Bit synchronization'**
- **Bit Synchronization is a function** that is required to determine when the **beginning** and **end of the data transmission** occurs.
- Bit synchronization **helps the receiving computer to know** when data begin and end during a transmission.
- Therefore bit synchronization provides **timing control**.



Asynchronous Transmission

- Asynchronous transmission sends only one character at a time where **a character is either a letter of the alphabet or number or control character** i.e. it sends one byte of data at a time.
- Bit synchronization between two devices is made possible using **start bit and stop bit.**
- Start bit indicates the **beginning of data** i.e. alerts the receiver to the arrival of new group of bits.
- A start bit usually **0** is added to the beginning of each byte.
- Stop bit indicates the end of data i.e. to let the receiver know that byte is finished, **one or more additional bits** are appended to the end of the byte. These bits, usually 1s are called stop bits.
- The time between sending and receiving data bits is not constant, so gaps are used to provide time between transmissions.
- The gap can be of varying intervals. This mechanism is called Asynchronous.

- Addition of start and stop increase the number of data bits.
- Hence more **bandwidth is consumed in asynchronous transmission**



Advantages of Asynchronous transmission

- This method of data transmission **is cheaper in cost as compared to synchronous** e.g. If **lines are short, asynchronous transmission is better**, because line cost would be low and idle time will not be expensive.
- In this approach each individual character is complete in itself, therefore if character is **corrupted during transmission**, its **successor and predecessor character** will not be affected.

Disadvantages of asynchronous transmission

- This method is less efficient and **slower than synchronous transmission** due to the overhead of extra bits and insertion of gaps into bit stream.
- Successful transmission inevitably depends on the recognition of the start bits. These bits can be missed or corrupted.

Example:

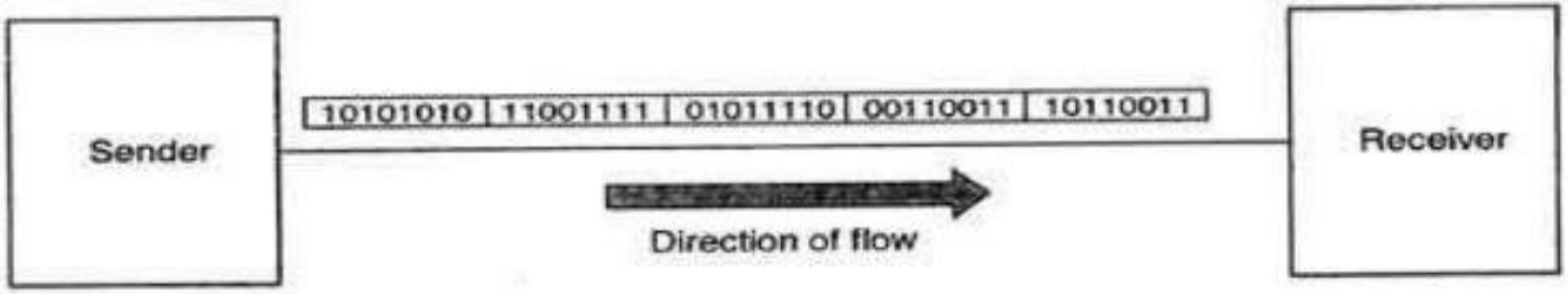
- Email
- Forums
- Letters

Application

- Asynchronous transmission is well suited for **keyboard type-terminals**. The advantage of this method is that it does not require any local storage at the terminal or the computer as transmission takes place character by character.

Synchronous Transmission

- Synchronous transmission **does not use start and stop bits**.
- In this method **bit stream is combined into longer frames** that may contain multiple bytes.
- There **is no gap between the various bytes** in the data stream.
- data moves faster and timing errors are less frequent because the transmitter and receiver time is synced



Synchronous Transmission

- In the absence of start & stop bits, bit synchronization is established between sender & receiver **by 'timing' the transmission of each bit**.
- Since the various bytes are placed on the link **without any gap**, it is the responsibility of **receiver to separate the bit stream into bytes so as to reconstruct the original information.**
- In order to receive the **data error free**, the receiver and sender operates at the same clock frequency.

Advantage of Synchronous transmission

- This method is **faster as compared to asynchronous** as there are no extra bits (start bit & stop bit) and also there is no gap between the individual data bytes.

Disadvantages of Synchronous transmission

- It is **costly** as compared to asynchronous method.
- It requires local buffer storage at the two ends of line to assemble blocks and it also requires accurately **synchronized clocks at both ends**.
- This lead to **increase in the cost**.
- The **sender and receiver have to operate at the same clock frequency**. This requires proper synchronization which makes the system complicated.

Examples

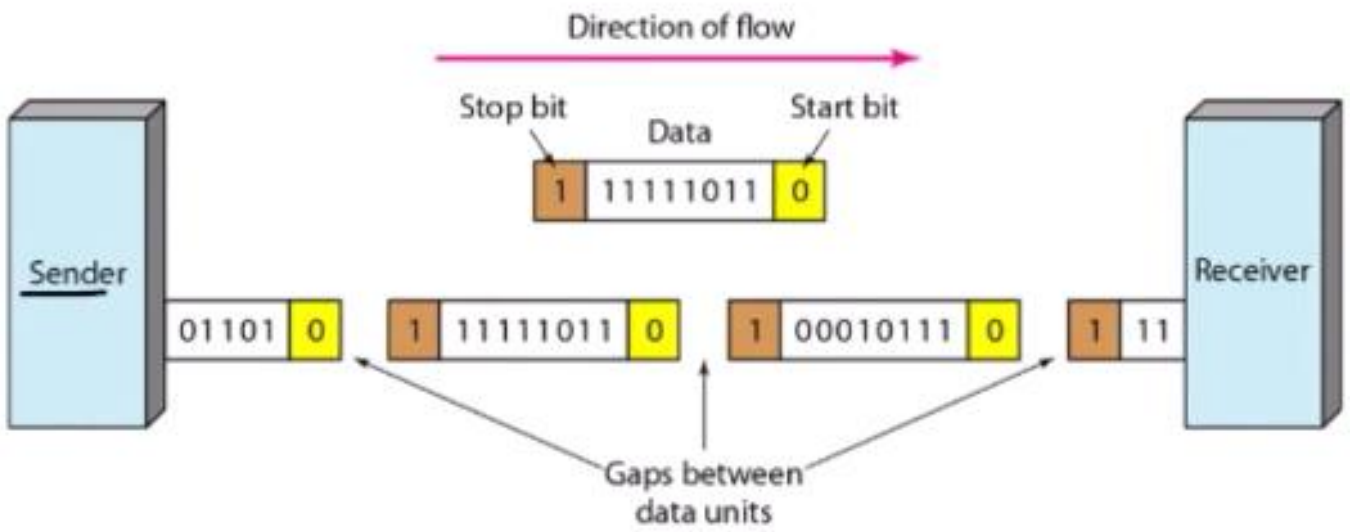
- Chat Rooms
- Telephonic Conversations
- Video Conferencing

Isochronous Transmission

- It uses both synchronous as well as asynchronous mode.(i.e. in synchronous timing is important (**synchronized**) but not in asynchronous . It is alike asynchronous where it has start and stop bit)
- Isochronous communication is a technique where **data is transmitted at regular intervals**, to maintain a **consistent data stream (Gaps between data are fixed)**

Timing between data frames is equal i.e Constant and regular time intervals

- **Data must arrive on time**, even if minor errors occur.
- **No retransmission is used (to prevent delays).**
- Supports **real-time data flow**



Advantages

- Isochronous delivers data at **regular, fixed time intervals**, which neither synchronous nor asynchronous
- Isochronous does **not retransmit data**, so data arrives quickly.

Synchronous = may wait for blocks

Asynchronous = has start/stop overhead

- This method is **faster as compared to asynchronous**

Disadvantages

- **No Retransmission** ,If a packet arrives corrupted.
- Wastes bandwidth if no data is available
- is **more complex** than simple asynchronous or synchronous systems.

Application

Isochronous is better for smooth real-time audio/video

When is serial transmission used to send data?

- Serial transmission is normally used for **long-distance data transfer**.
- It is also used in cases where the amount of data being sent **is relatively small**.
- It ensures that data integrity is maintained as it transmits the data bits in a specific order, one after another.
- In this way, data bits are received in-sync with one another.

Difference Between Asynchronous and Synchronous Transmission

BASIS FOR COMPARISON	SYNCHRONOUS TRANSMISSION	ASYNCHRONOUS TRANSMISSION
Meaning	Transmission starts with the block header which holds a sequence of bits.	It uses start bit and stop bit preceding and following a character respectively.
Transmission manner	Sends data in the form of blocks or frames	Sends 1 byte or character at a time
Synchronization	Present with the same clock pulse.	Absent
Gap between the data	Does not exist	Exist
Transmission Speed	Fast	Speed
Cost	Expensive	Economical

Comparison between Serial and Parallel transmission

Serial Transmission	Parallel Transmission
Bit by Bit transmission & each bit requires one clock cycle.	Multiple bits are transmitted in one clock cycle.
1 bit in one clock cycle	8 bit in one clock cycle
Economical	Expensive
Slow	Fast
Used for long distance communication	Used for short distance communication
Only one communication channel required	N number of communication channel required
Communication between computers in remote areas.	Communication between computer and printer.

Modes of Data Transmission

- Data transmission modes describe **how data flows** between two communicating devices
- There are **three main modes**:

  Simplex – One-way communication only.

  Half-Duplex – Two-way communication but one direction at a time.

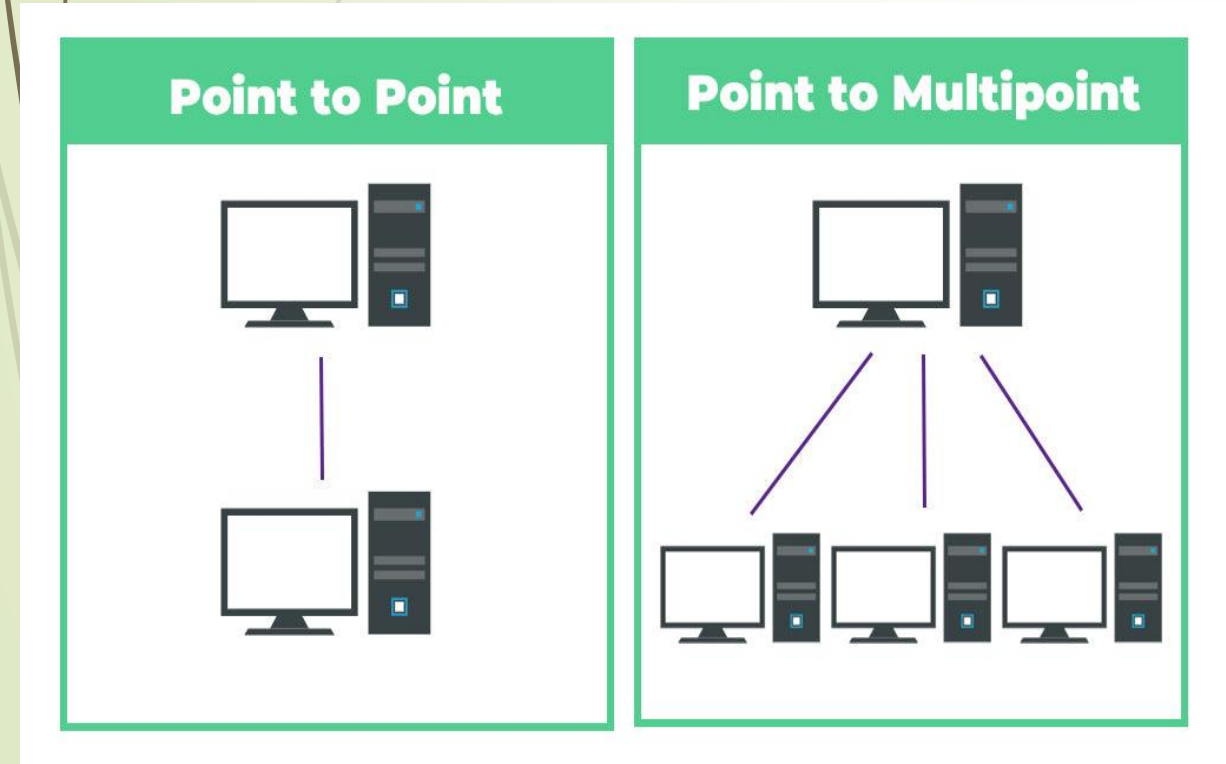
  Full-Duplex – Two-way simultaneous communication.

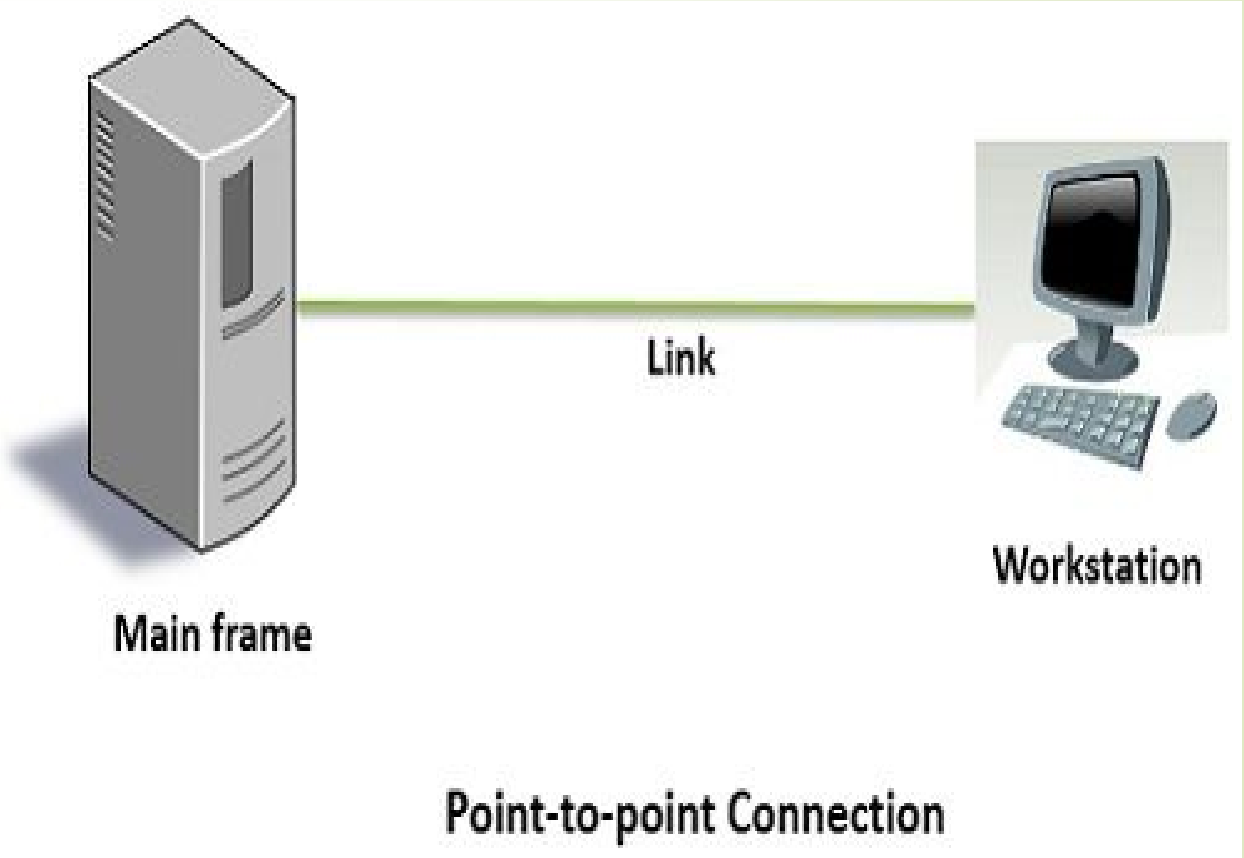
Line Configuration

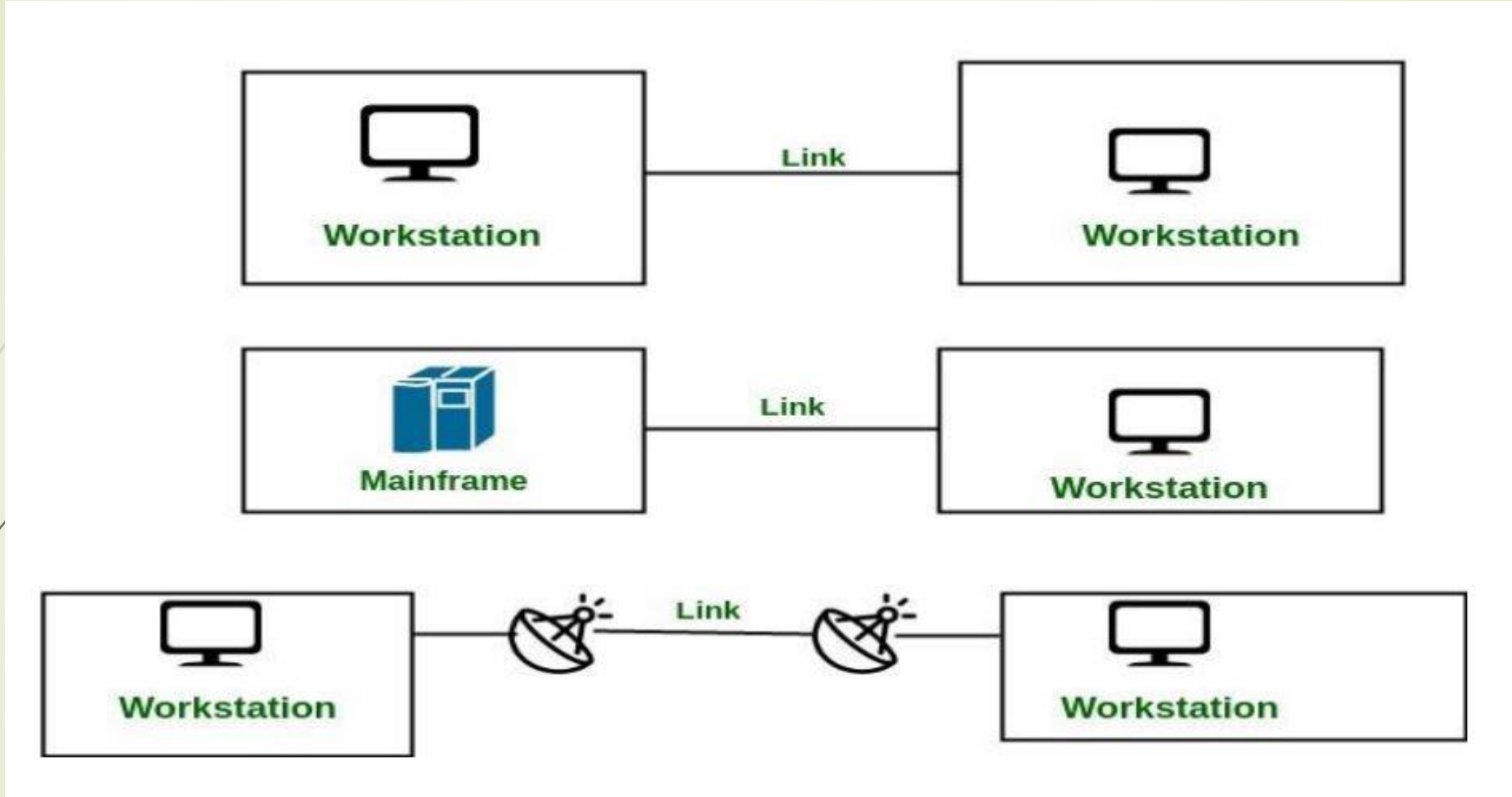
- It refers to the way through which **two or more communication devices** attached to a link.
- Devices can be a computer, printer or any other device that is capable to send and receive data.
- There are two ways to connect the devices :
- **Point-to-Point connection**
- **Multipoint connection**

Point-to-Point Connection

- A point-to-point connection provides **a dedicated link** between two devices.
- The **entire capacity of the link is reserved** for transmission between those two devices.
- Mostly **cables/wires** are used to attach the two endpoints in **point-to-point line configuration**, but other options such as microwave or satellite links are also possible.
- Example: Point-to-Point connection between **remote control and Television** for changing the channels.






Workstation
Link
Workstation
Mainframe
Link
Workstation
Workstation
Link
Workstation

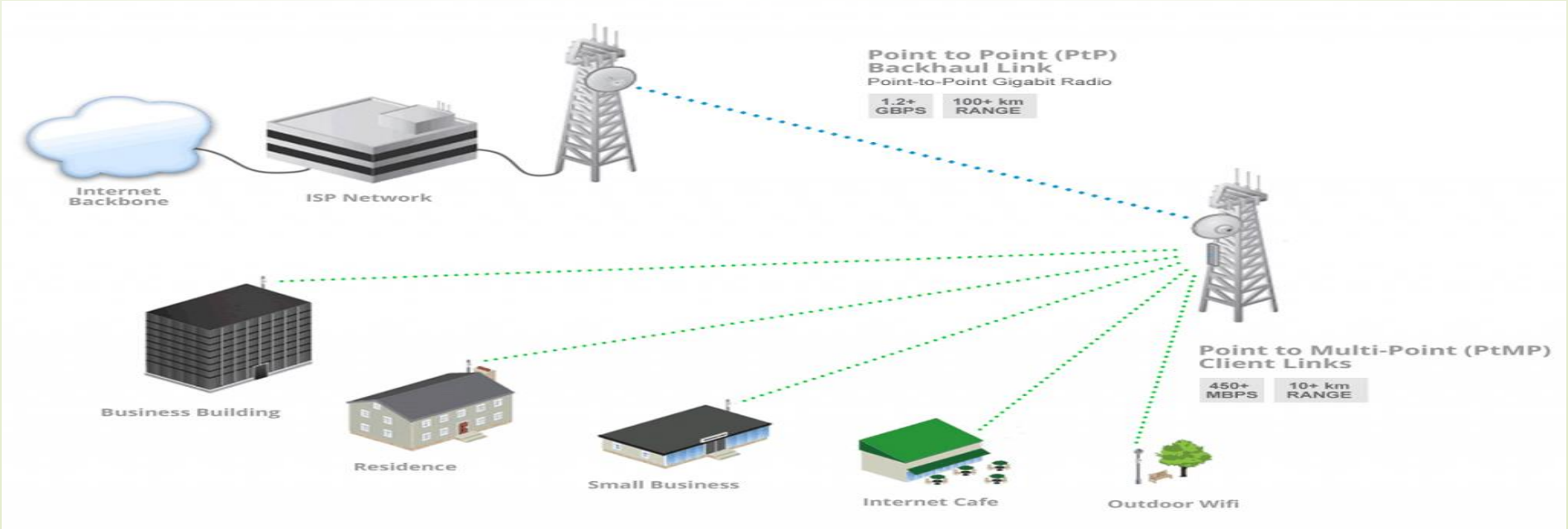
Multipoint Connection:

- It is also called **Multidrop configuration**.
- In this connection **two or more devices share a single link**.

More than two devices share the link that is the capacity of the channel is shared now.

- With shared capacity, there can be **two possibilities** in a Multipoint Line configuration:

- **Spatial Sharing:** If several devices can share the link **simultaneously**, its called Spatially shared line configuration.
- **Temporal (Time) Sharing:** If users must take **turns using the link** , then its called Temporally shared or **Time Shared Line configuration**



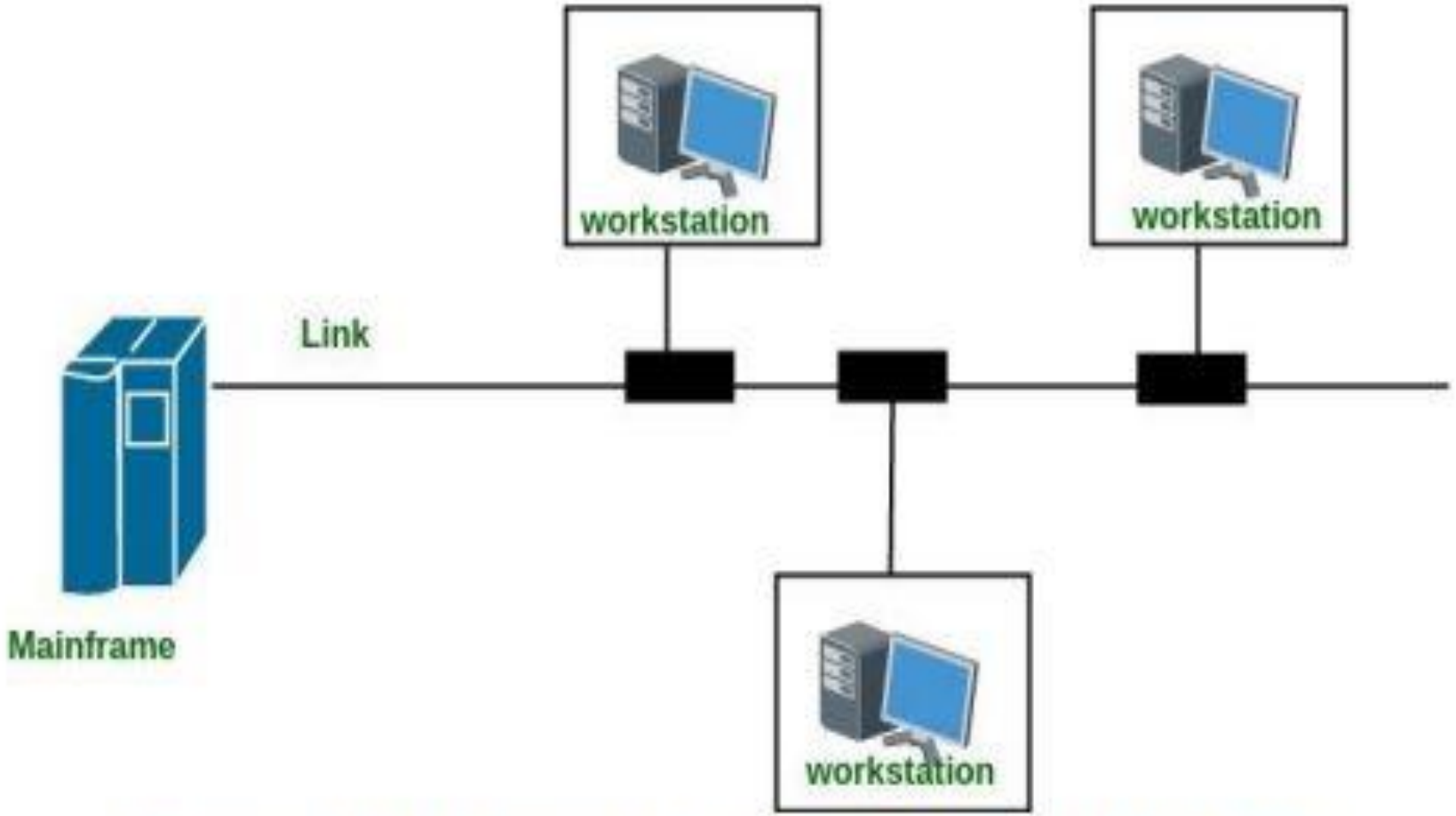
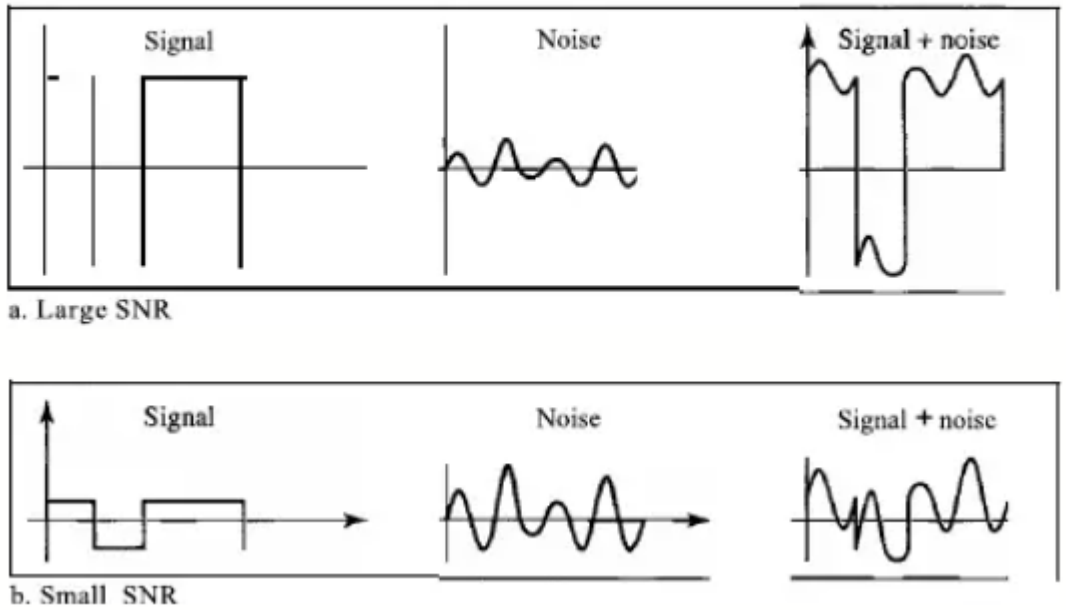


Figure 2.9: Multipoint Protocol (Temporal Sharing)

Figure 3.30 *Two cases of SNR: a high SNR and a low SNR*



3.5 DATA RATE LIMITS

A very important consideration in data communications is how fast we can send data, in bits per second. over a channel. Data rate depends on three factors:

1. The bandwidth available
2. The level of the signals we use
3. The quality of the channel (the level of noise)

Two theoretical formulas were developed to calculate the data rate: one by Nyquist for a noiseless channel, another by Shannon for a noisy channel.

Solution

The values of SNR and SNR_{dB} can be calculated as follows:

$$SNR = \frac{10{,}000\ \mu W}{1\ mW} = 10{,}000$$

$$SNR_{dB} = 10 \log_{10} 10{,}000 = 10 \log_{10} 10^4 = 40$$

Example 3.32

The values of SNR and SNR_{dB} for a noiseless channel are

$$SNR = \frac{\text{signal power}}{0} = \infty$$

$$SNR_{dB} = 10 \log_{10} \infty = \infty$$

We can never achieve this ratio in real life; it is an ideal.

Noiseless Channel: Nyquist Bit Rate

For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate

$$\text{BitRate} = 2 \times \text{bandwidth} \times \text{10g2}\ L$$

In this formula, bandwidth is the bandwidth of the channel, L is the number of signal levels used to represent data, and BitRate is the bit rate in bits per second.

According to the formula, we might think that, given a specific bandwidth, we can have any bit rate we want by increasing the number of signal levels. Although the idea is theoretically correct, practically there is a limit. When we increase the number of signal levels, we impose a burden on the receiver. If the number of levels in a signal is just 2, the receiver can easily distinguish between a 0 and a 1. If the level of a signal is 64, the receiver must be very sophisticated to distinguish between 64 different levels. In other words, increasing the levels of a signal reduces the reliability of the system.

Increasing the levels of a signal may reduce the reliability of the system.

Example 3.34

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$

Example 3.35

Consider the same noiseless channel transmitting a signal with four signal levels (for each level, we send 2 bits). The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 4 = 12{,}000 \text{ bps}$$

Example 3.36

We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

Solution

We can use the Nyquist formula as shown:

$$265{,}000 = 2 \times 20{,}000 \times \log_2 L$$

$$\log_2 L = 6.625 \qquad L = 2^{6.625} = 98.7 \text{ levels}$$

Since this result is not a power of 2, we need to either increase the number of levels or reduce the bit rate. If we have 128 levels, the bit rate is 280 kbps. If we have 64 levels, the bit rate is 240 kbps.

Noisy Channel: Shannon Capacity

In reality, we cannot have a noiseless channel; the channel is always noisy. In 1944, Claude Shannon introduced a formula, called the Shannon capacity, to determine the theoretical highest data rate for a noisy channel:

$$\text{Capacity} = \text{bandwidth} \times \log_2 (1 + \text{SNR})$$

In this formula, bandwidth is the bandwidth of the channel, SNR is the signal-to-noise ratio, and capacity is the capacity of the channel in bits per second. Note that in the Shannon formula there is no indication of the signal level, which means that no matter how many levels we have, we cannot achieve a data rate higher than the capacity of the channel. In other words, the formula defines a characteristic of the channel, not the method of transmission.

1.Channel Capacity

The channel capacity, C, is defined to be the maximum rate at which information can be transmitted through a channel.

The Shannon capacity theorem defines the maximum amount of information, or data capacity, which can be sent over any channel or medium (wireless, coax, twisted pair, fiber etc.).

bandwidth of the channel

$$C = B \log_2 (1+S/N)$$

Channel capacity in bits/s

signal-to-noise ratio

where

C is the channel capacity in bits per second (or maximum rate of data)

B is the bandwidth in Hz available for data transmission

S is the received signal power

N is the total channel noise power across bandwidth B

Higher the signal-to-noise (SNR) ratio and more the channel bandwidth, the higher the possible data rate.

This equation sets the theoretical upper limit on data rate, which of course is not fully achieved in practice.

Example 3.37

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as

$$C = B \log_2 (1 + SNR) = B \log_2 (1 + 0) = B \log_2 1 = B \times 0 = 0$$

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.

Example 3.38

We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as

$$C = B \log_2 (1 + SNR) = 3000 \log_2 (1 + 3162) = 3000 \log_2 3163$$
$$= 3000 \times 11.62 = 34{,}860 \text{ bps}$$

This means that the highest bit rate for a telephone line is 34.860 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.

Example 3.39

The signal-to-noise ratio is often given in decibels. Assume that $SNR_{dB} = 36$ and the channel bandwidth is 2 MHz. The theoretical channel capacity can be calculated as

$$SNR_{dB} = 10 \log10 SNR \rightarrow SNR = 10^{SNR_{dB}/10} \rightarrow SNR = 10^{3.6} = 3981$$

$$C = B \log2 (1 + SNR) = 2 \times 10^6 \times \log2 3982 = 24 \text{ Mbps}$$

Example 3.40

For practical purposes, when the SNR is very high, we can assume that SNR + I is almost the same as SNR. In these cases, the theoretical channel capacity can be simplified to

$$C = B \times \frac{SNR_{dB}}{3}$$

For example, we can calculate the theoretical capacity of the previous example as

$$C = 2 \text{ MHz} \times \frac{36}{3} = 24 \text{ Mbps}$$

Using **Both** Limits

In practice, we need to use both methods to find the limits and signal levels. Let us show this with an example.

Example 3.41

We have a channel with a I-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Solution

First, we use the Shannon formula to find the upper limit.

$$C = B \log_2 (1 + SNR) = 10^6 \log_2 (1 + 63) = 10^6 \log_2 64 = 6 \text{ Mbps}$$

The Shannon formula gives us 6 Mbps, the upper limit. For better performance we choose something lower, 4 Mbps, for example. Then we use the Nyquist formula to find the number of signal levels.

$$4\text{Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L \quad \Rightarrow \quad L = 4$$

The Shannon capacity gives us the upper limit;
the Nyquist formula tells us how many signal levels we need.

2.3 Bit Rate/ Baud Rate, Transmission Channel, RS-232C and RS-449 Interface Standards

- **Bit Rate and Baud Rate in data transmission**
- **Bit rate**
- It is the transmission of number of bits per second.
- It determines the number of bits traveled per second.
- **Bit rate = baud rate x the number of bits per signal unit**

- **Baud Rate:**

✓ Baud rate is expressed in the number of times a signal can change on transmission line per second.

For example

- 1500 baud rate illustrates that the channel state can alter up to 1500 times per second. The meaning of changing state means that channel can change its state from 0 to 1 or from 1 to 0 up to 1500 times per second
- **Baud rate = bit rate / the number of bits per signal unit**

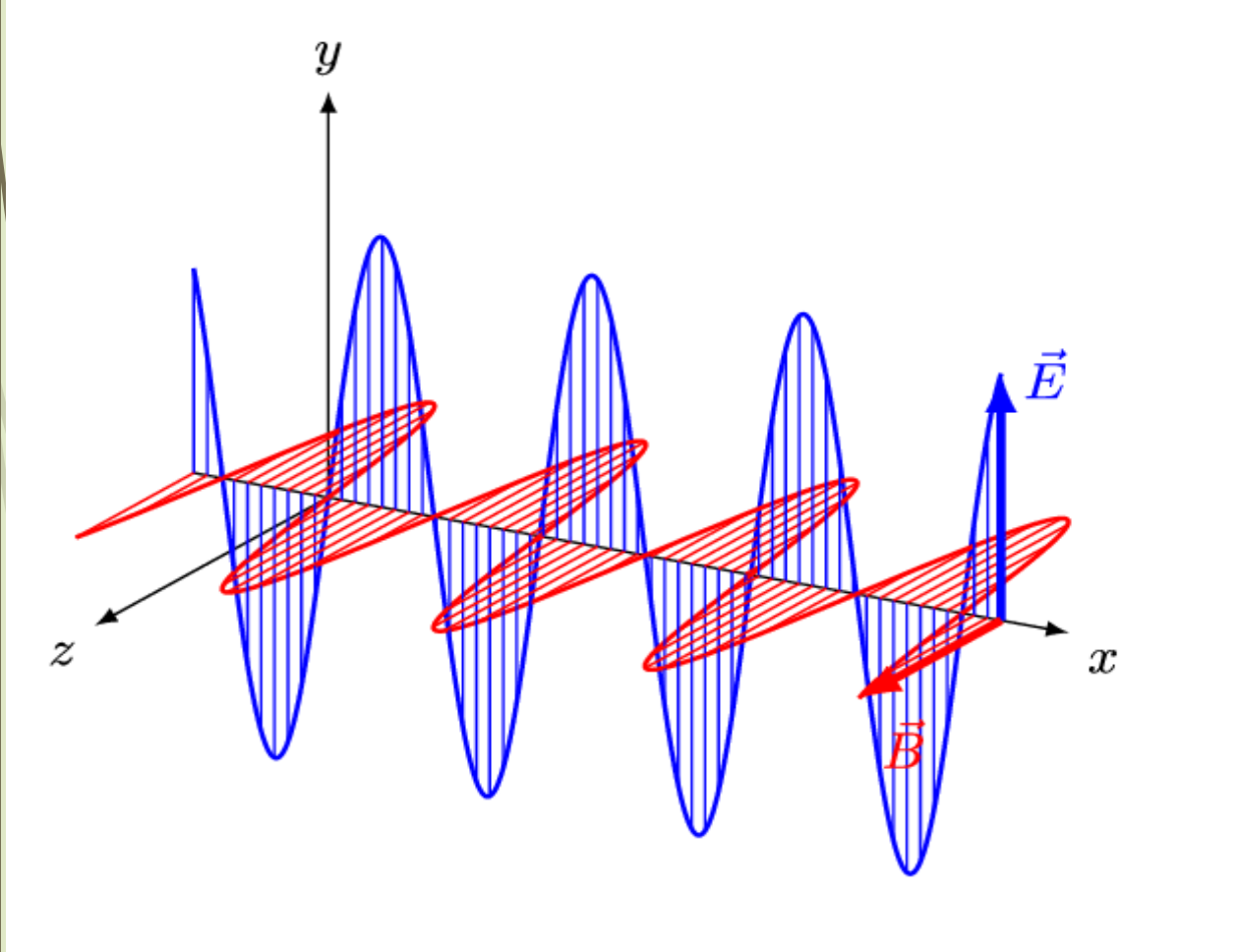
Difference between bit rate and baud rate

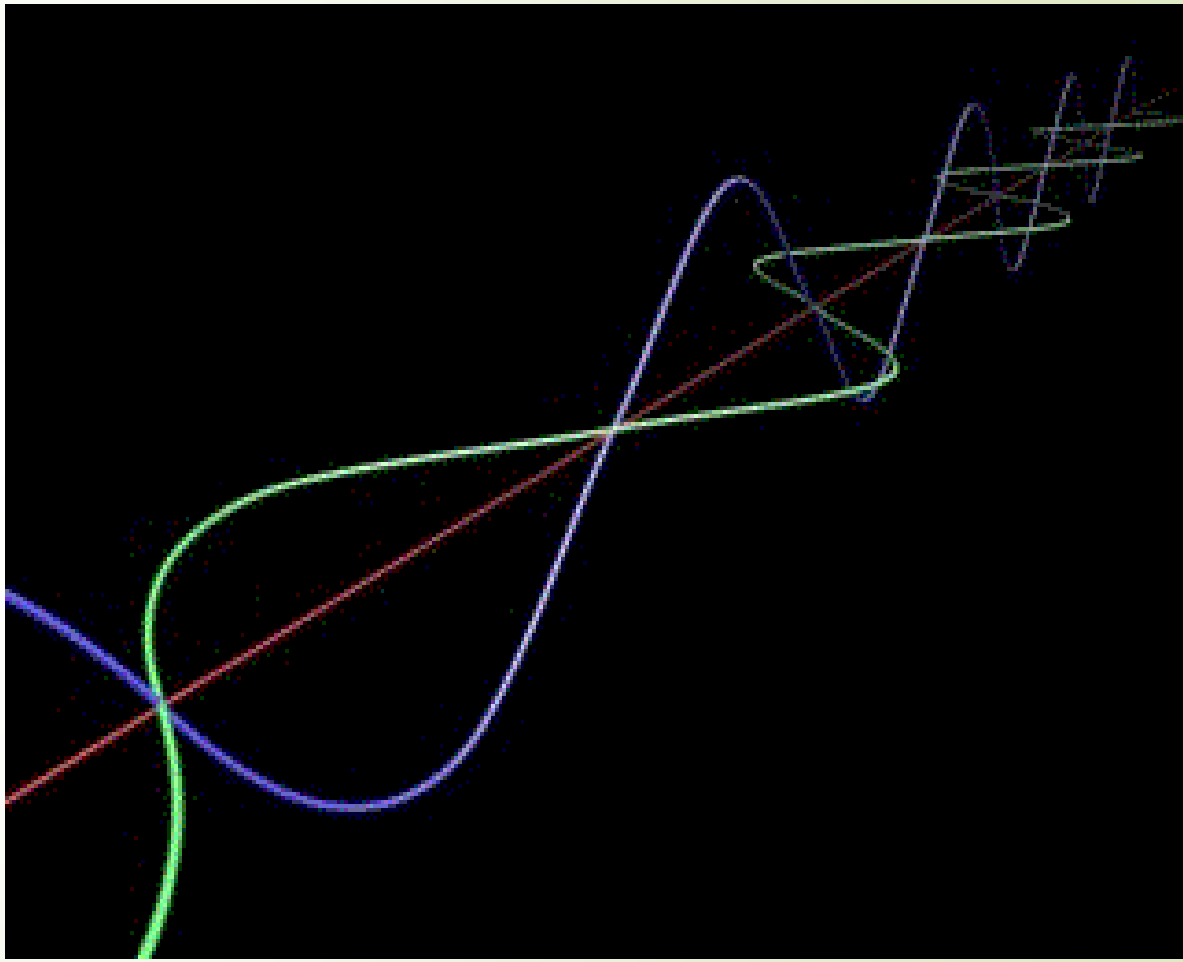
- 1. Bit rate is defined as the transmission of number of bits per second where as **Baud rate** is defined as the number of signal units per second.
- 2. **Bit rate** is also defined as per second travel number of bits where as **Baud rate** is also defined as per second number of changes in signal.
- 3**. Bit rate** emphasized on computer efficiency While **baud rate** emphasized on data transmission.
- 4. The formula of Bit Rate is:- **baud rate x the number of bit per baud**
- 5. The formula of Baud Rate is:- **bit rate / the number of bit per baud**

Transmission Media

- Transmission media is a **communication channel that carries the information** from the sender to the receiver.
- Data is transmitted through the **electromagnetic signals**.
- The characteristics and quality of data transmission are determined by the characteristics of medium and signal.
- Transmission media is of two types are wired media and wireless media.
- In wired media, **medium characteristics** are more important whereas, in wireless media, **signal characteristics** are more important.

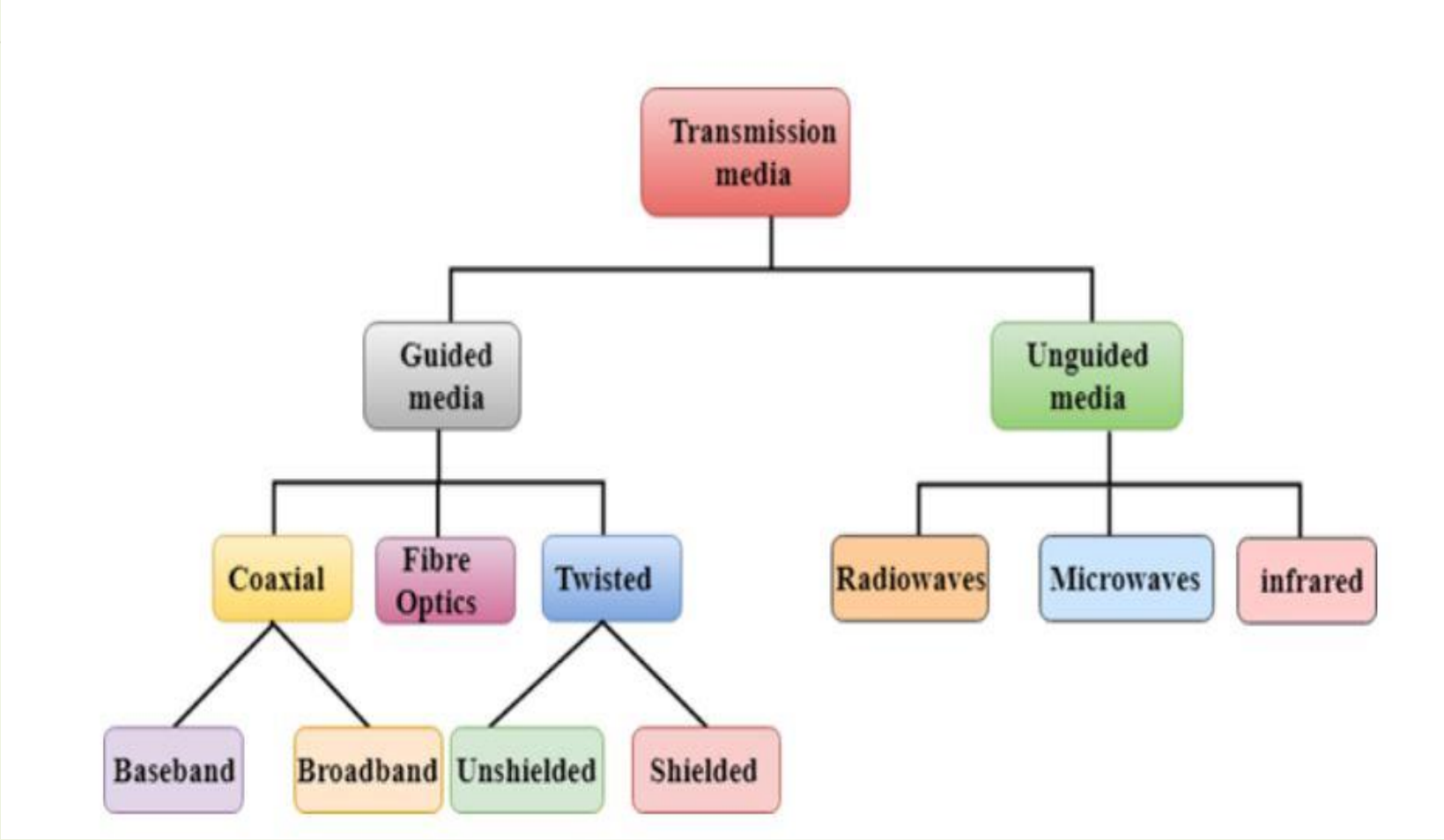
EM wave





Some factors need to be considered for designing the transmission media:

- **Bandwidth**: All the factors are remaining constant, **the greater the bandwidth of a medium, the higher the data transmission rate of a signal.**
- **Transmission impairment:** When the received signal is not identical to the transmitted one due to the transmission impairment. **The quality of the signals will get destroyed due to transmission impairment.**
- **Interference:** An interference is defined as the process of **disrupting a signal** when it travels over a communication medium on the **addition of some unwanted signal**.


Transmission media
Guided media
Unguided media
Coaxial
Fibre Optics
Twisted
Radiowaves
Microwaves
infrared
Baseband
Broadband
Unshielded
Shielded

RS232C(RS-232C (DTE-DCE, DTE-DTE))

- RS-232 is a standard communication protocol for connecting computers and their peripheral devices to enable serial data exchange.
- RS-232C is the interface that your computer uses to talk to and exchange data with your modem and other serial devices.
- The *RS* in RS-232C stands for "Recommended Standard," and the *C* refers to the version.

RS-232C itself describes the physical and electrical characteristics of serial communication for data transfer.

- For connecting **Data Terminal Equipment (DTE)** and **Data Communication Equipment (DCE)**.

- **Type:** Asynchronous serial communication
- **Distance:** Up to **15 meters (50 feet)**
- **Speed:** Typically **up to 20 kbps**

Voltage levels:

•Logic 1 (MARK) → **–3V to –15V**

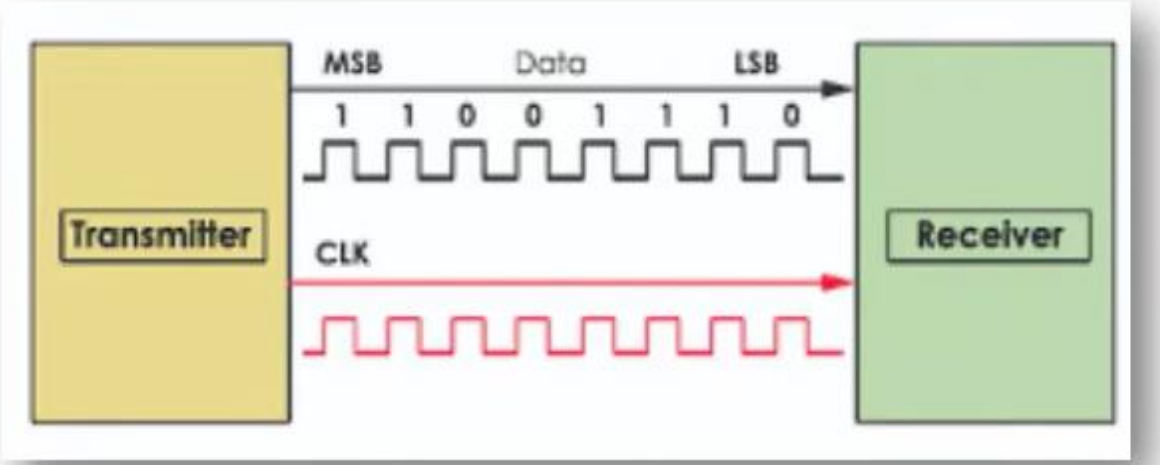
•Logic 0 (SPACE) → **+3V to +15V**

•Negative voltage represents binary 1; positive voltage represents binary 0.

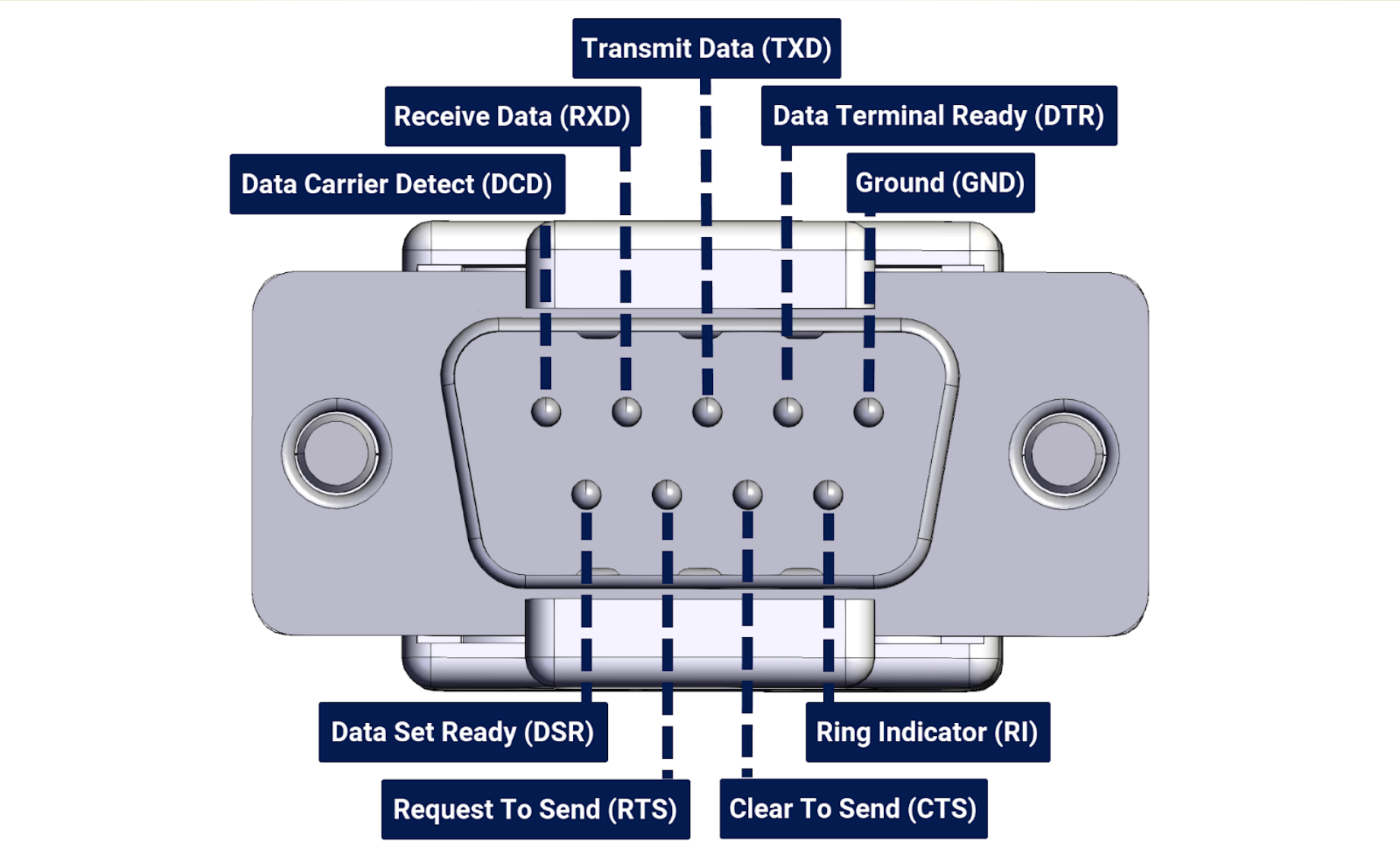
•Commonly used with **DB-25** or **DB-9 connectors**

What is RS 232?

- RS232C **"Recommended Standard 232C"**.
- In new PC's having **D-type connector which is of 9 pins**.
- RS232 is a standard protocol used for serial communication.
- It is used for connecting computer and its peripheral devices to allow serial data exchange between them.
- It is used in serial communication with the rate of 1.492 kbps.



DB9 pinout


Transmit Data (TXD)
Receive Data (RXD)
Data Terminal Ready (DTR)
Data Carrier Detect (DCD)
Ground (GND)
Data Set Ready (DSR)
Ring Indicator (RI)
Request To Send (RTS)
Clear To Send (CTS)

RS-232 Connector

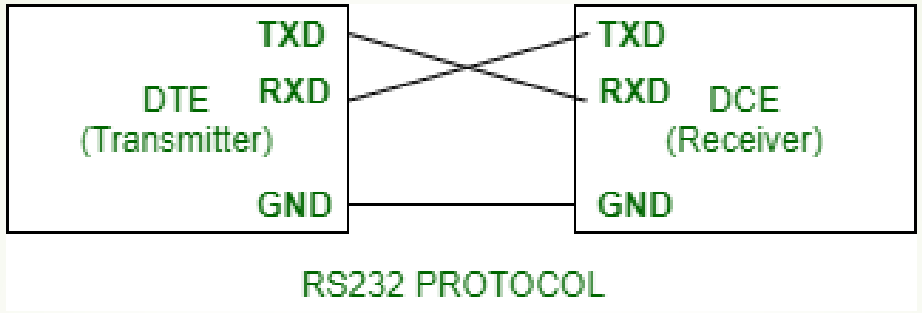
- The RS-232 standard **employs DB9 and DB25 connectors**, each characterized by its **D-shaped** form. These connectors **facilitate communication with their male and female** interfaces on either end.

DB9 connector is a **D-subminiature (D-Sub) connector with a "B" shell size and 9 pins**

"Data Bus": In the "DB" part of the name, "D" stands for the D-shaped shell, and the "B" (or "E" in the correct term) refers to the shell size.

https://www.solisplc.com/tutorials/rs232#:~:text=To%20connect%20two%20DTE%20devices%20directly%2C%20without,2%20and%20Pin%203%20at%20each%20end.

connection with RS232 for transmitting data to any serial device like modem or printer from its DTE interface.



https://www.geeksforgeeks.org/computer-networks/rs232c-for-data-transfer/

DTE (Data Terminal Equipment)

- End device that generates or receives data.
- Example: **Computer, terminal, printer, microcontroller**
- Usually **male connector** in RS-232C
- Typically uses **Transmit Data (TXD)** to send and **Receive Data (RXD)** to receive.

DCE (Data Communication Equipment)

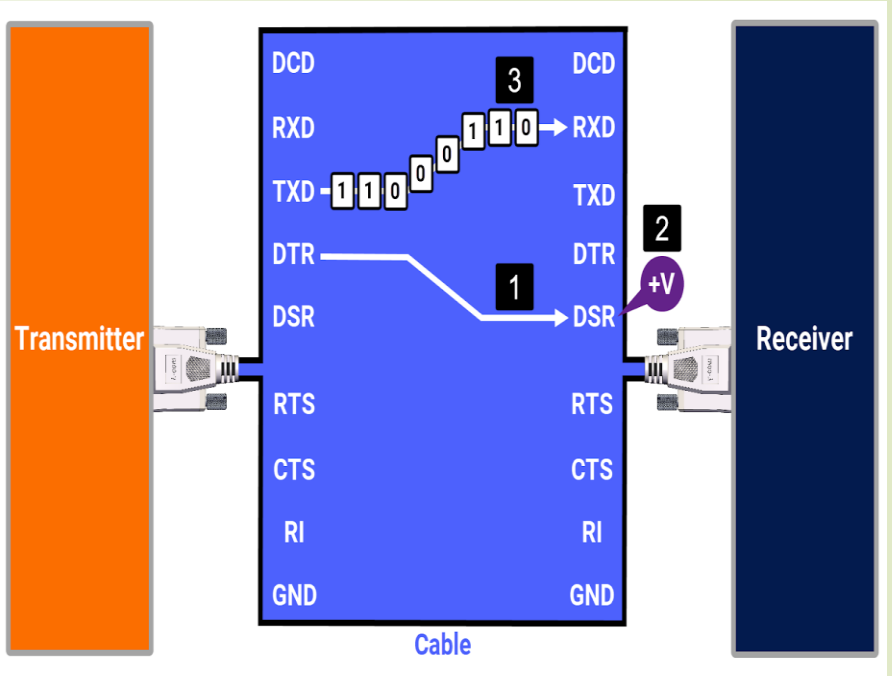
- Device that provides a communication link between DTE and network.
- Example: **Modem, router**
- Usually **female connector** in RS-232C
- Its TXD and RXD are **opposite roles** of DTE

DTE–DCE Connection (Standard Connection)

- This is the **normal** and intended connection in RS-232C.

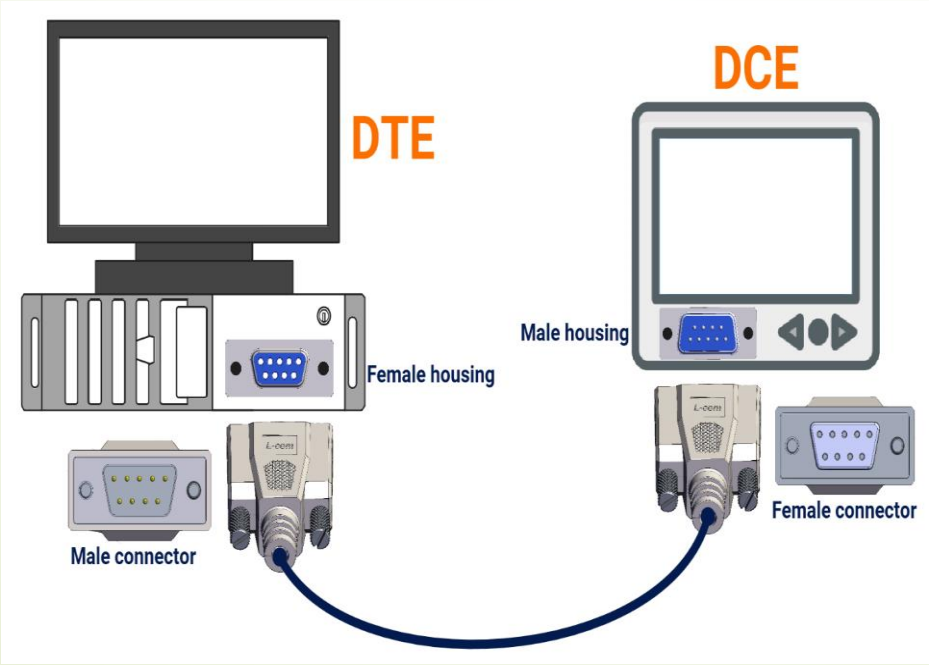
Characteristics

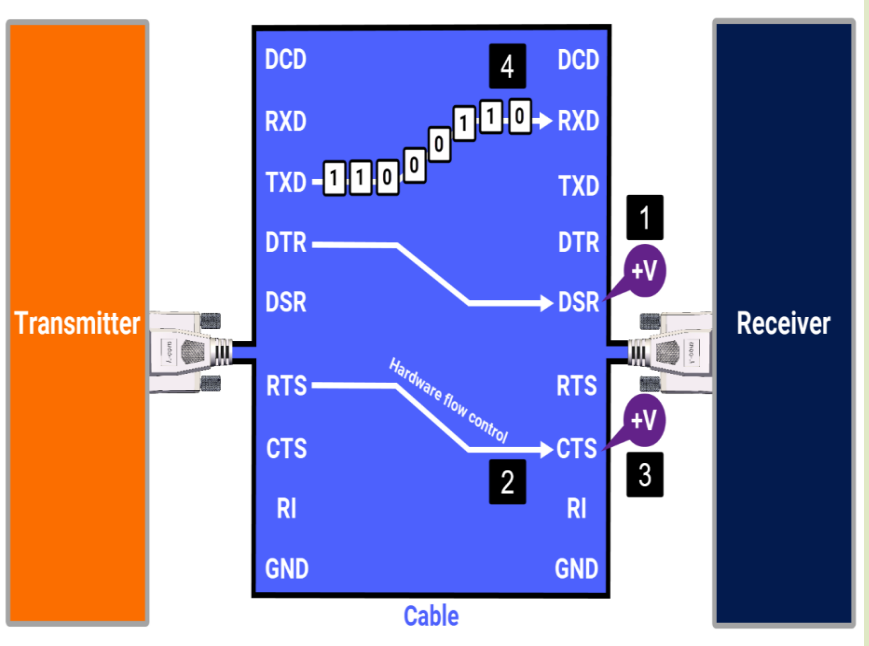
- Straight-through cable (TXD from DTE → RXD of DCE)
- No crossover required
- Most serial communication standards are designed for this pair



Pin Mapping (Straight-through)

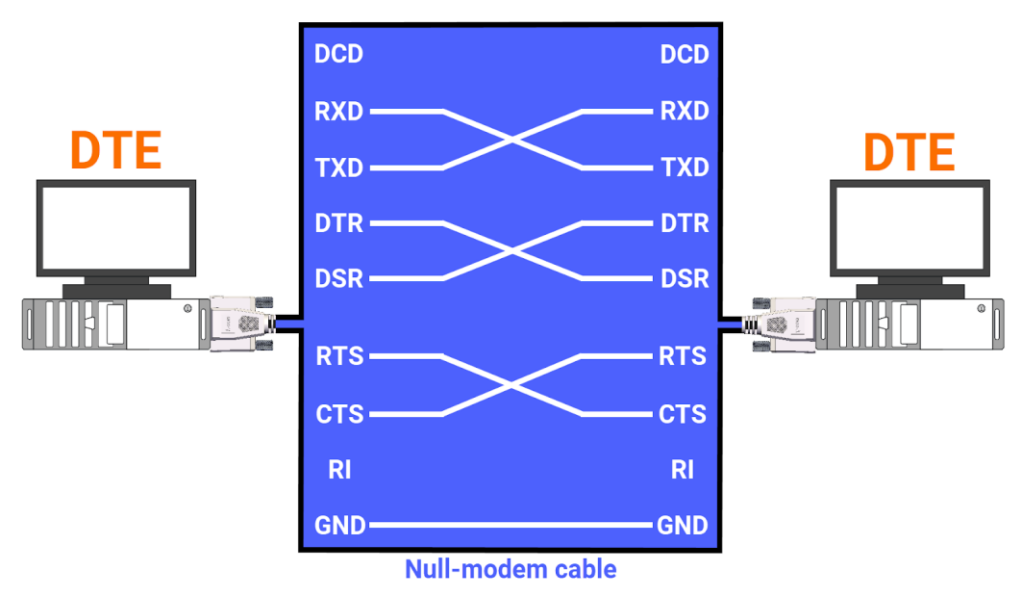
- **DTE TXD → DCE RXD**
- **DTE RXD → DCE TXD**
- **DTE RTS → DCE CTS**
- **DTE DTR → DCE DSR**





DTE–DTE Connection (Crossover Connection)

- This is used when connecting **two computers directly** without a modem or DCE device.



Characteristics

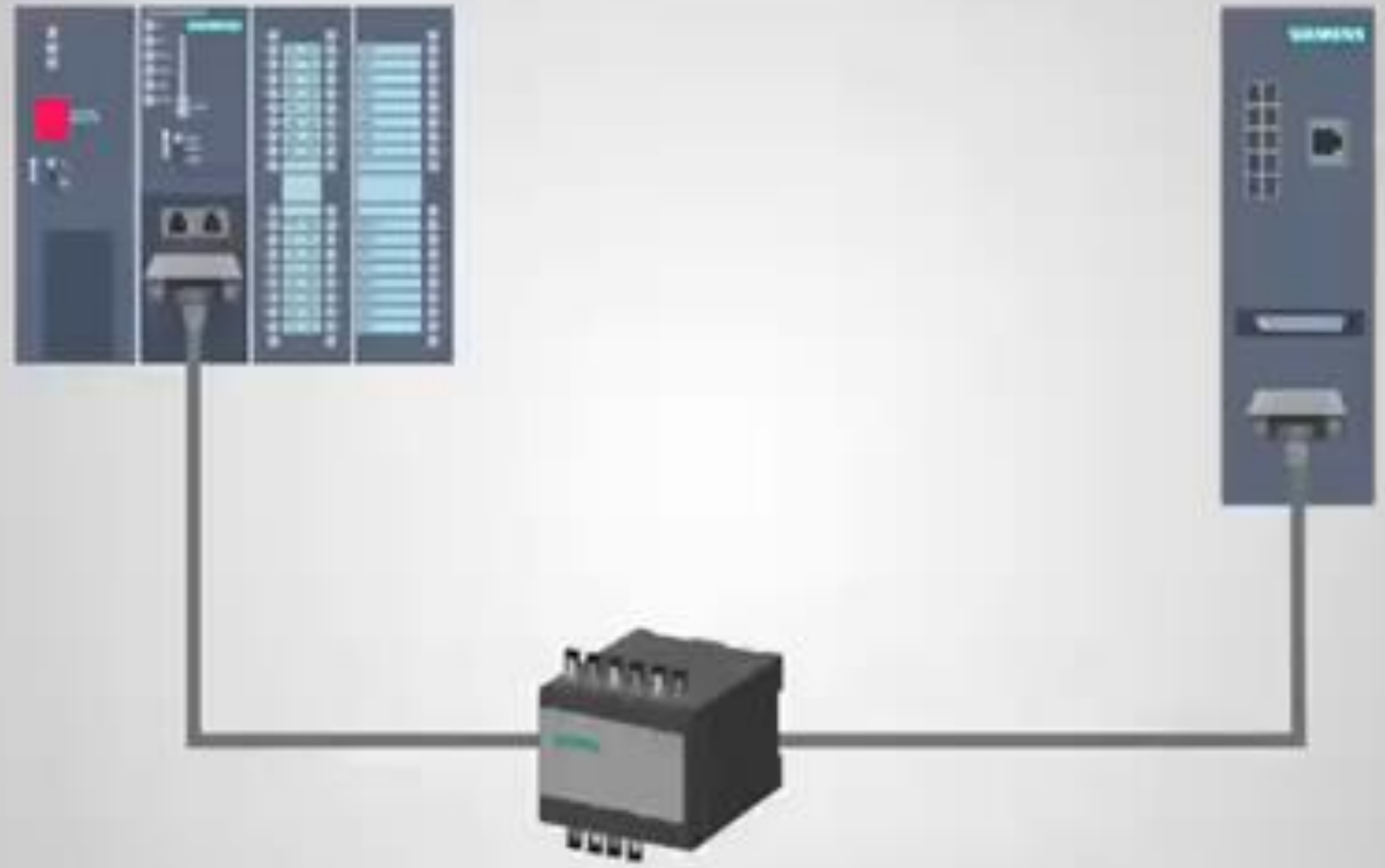
- Requires a **null-modem cable (crossed cable)**
- TXD and RXD lines must be **crossed**
- Handshake signals also crossed

Pin Mapping (Crossed / Null-modem)

- **DTE1 TXD → DTE2 RXD**
- **DTE1 RXD → DTE2 TXD**
- **DTE1 RTS → DTE2 CTS**
- **DTE1 CTS → DTE2 RTS**
- Grounds common

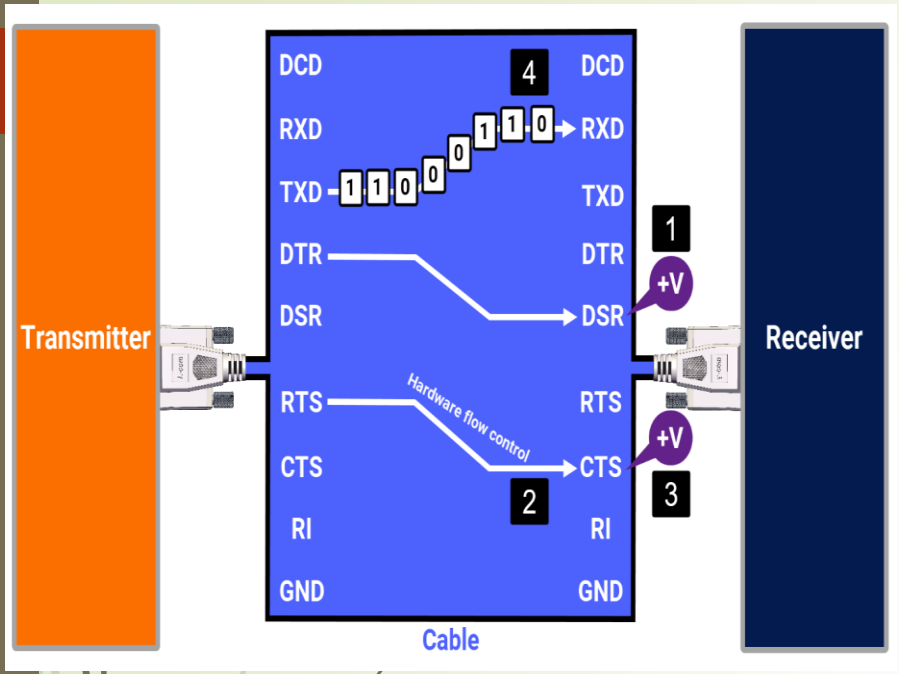
Use Case

- Direct file transfer between two PCs
- Microcontroller-to-PC direct programming/debugging
- Serial communication between two terminals without a modem



REALPARS

Extra



3. Voltage Levels (+V)

•RS-232 uses **positive and negative voltages** to represent logic levels.

•The diagram indicates +V near certain pins, showing that:

- **Positive voltage** = Logic 0 **(SPACE)**
- **Negative voltage** = Logic 1 (**MARK)**

1. Data Signals (RXD, TXD)

- **TXD (Transmit Data):**
  The transmitter sends serial bits (1s and 0s) to the receiver.
  In the diagram, the sequence **1 1 0 0 0 1 1 0** represents digital data being transmitted.
- **RXD (Receive Data):**
  The receiver reads this bit stream on the RXD line.
- These two lines carry the **actual data** in asynchronous serial communication.

2. Control Signals (RTS, CTS, DTR, DSR)

- These signals manage communication and ensure proper flow of data.
- **RTS (Request to Send) → CTS (Clear to Send)**
- Part of **hardware flow control**.
- The transmitter raises **RTS** when it wants to send data.
- The receiver responds with **CTS** when it is ready to receive.
- Shown in the diagram by arrows labeled *Hardware flow control*.
- This prevents data loss.

DTR (Data Terminal Ready) → DSR (Data Set Ready)

- DTR indicates the transmitter is powered on and ready.
- DSR shows the receiver/modem is ready.
- These maintain the **link status**.

4. DCD (Data Carrier Detect)

•Indicates that a valid communication signal (carrier) is detected.

•Used mainly in modem connections.